



Smart Pill Dispenser

6710110021

6710110562

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240-

319 Embedded System Developer Module

ESP32

IR

Wi-Fi

25

. . 2569

	1			1
1.1		1.2	1.3	1
	2			2
2.1	2.4			2
2.5	2.7			3
	3			4
3.1				4
3.2				5
3.3				6
3.4				7
3.5				8
3.6				9
	4			10
4.1				10
4.2				11
4.3				12
	5			13
				14

1

1.1

ESP32

IR

1.2

1.
ESP32

2.

3. IR

4.

1.3

1-9

2

2.1

ESP32

ESP32

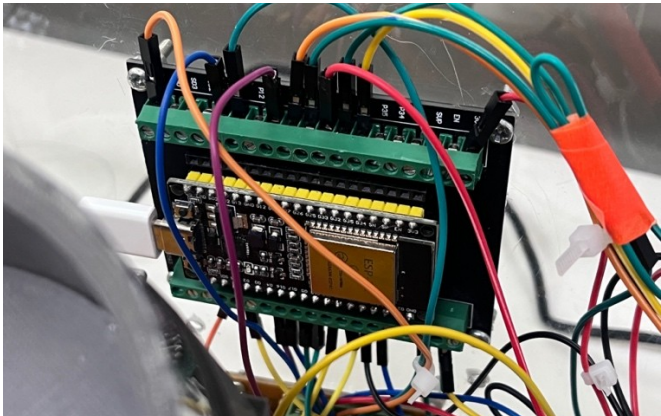
Wi-Fi

GPIO

ESP32 DevKit

ESP32-

WROOM-32



2.2

DS1307

LCD

DS1307

I²C

LCD

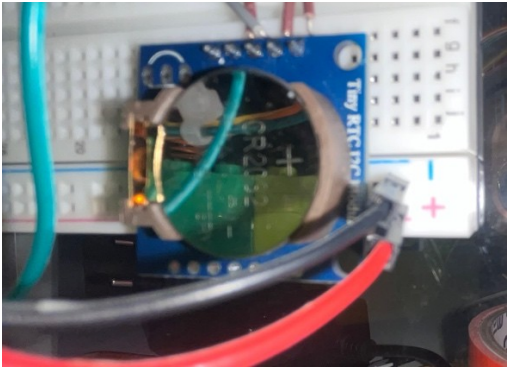
16×2

20×4

I²C

0x27

0x25



2.3

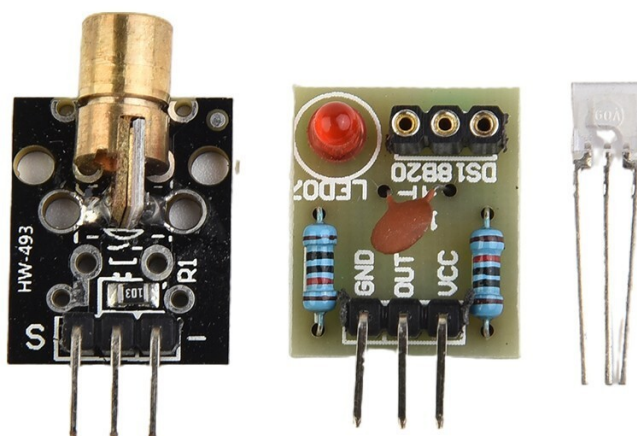
DRV8833

2.4

IR

IR

60



2

2.5 PI

ESP32

60

GET /api/device/sync

POST

/api/device/events

URL

HTTP

HTTPS

HTTPS

config.h

API key

JWT

2.6

React

TypeScript

NestJS

Prisma

PostgreSQL

account

2.7

loop

Flash/NVS

website

(Command ID)

3

3.1

1	ESP32 DevKit / WROOM-32	1	
2	RTC DS1307	1	
3	LCD I ² C 16×2	1	
4	LCD I ² C 20×4	1	
5		3	
6		3	
7	DRV8833	2	
8	IR	3	
9		1	
10	LED	1	
11	Buzzer	1	
12	I ² C	1	
13		1	
14		1	

15			

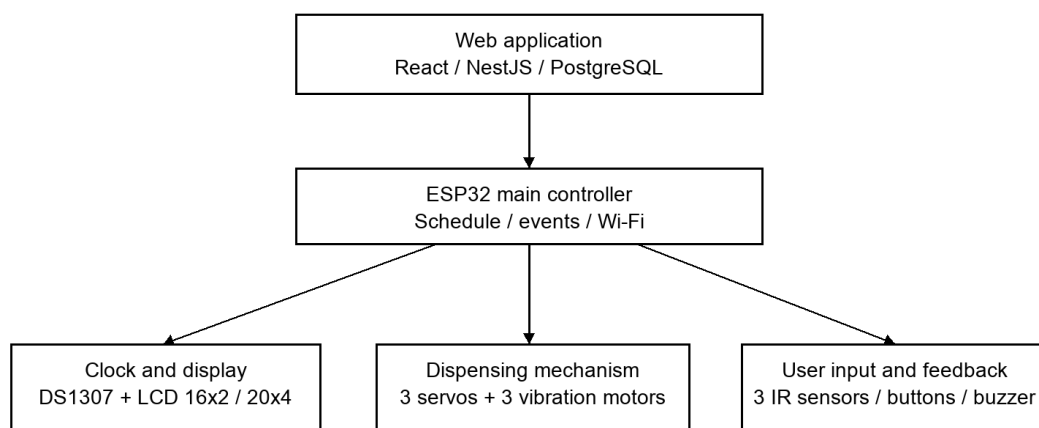
Arduino IDE

ESP32 ESP32Servo, ArduinoJson, LiquidCrystal_I2C, TimeLib

DS1307RTC Python C++

ESP32

3.2



1

		GPIO	ESP32
I ² C SDA	SCL	21	22
Servo	1 / 2 / 3	18 / 19 / 23	
	PWM	1 / 2 / 3	13 / 26 / 4
	DIR	1 / 2 / 3	16 / 17 / 14
IR	1 / 2 / 3	34 / 35 / 36	
	/	/	33 / 32 / 27
Buzzer		25	

RTC

I²C

IR

ESP32

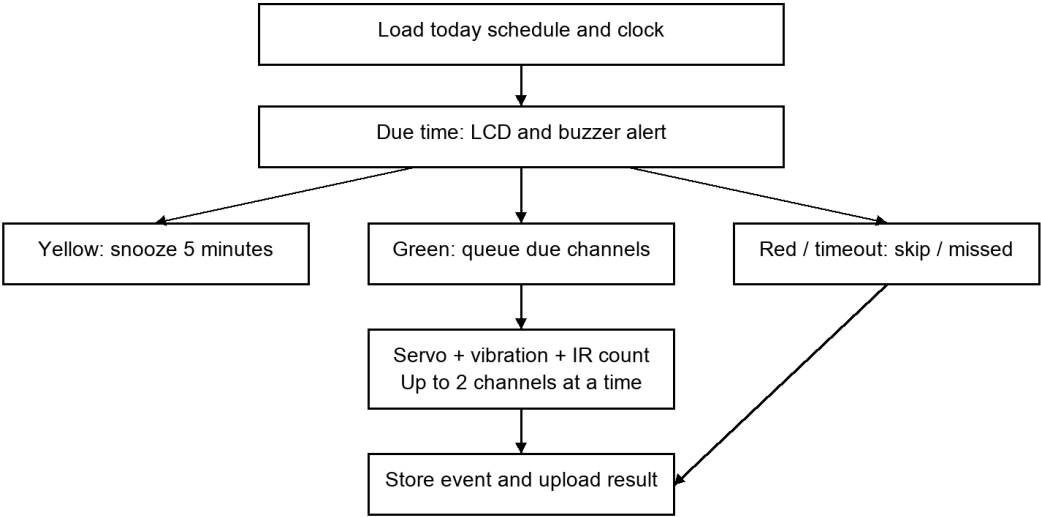
GND

ESP32

3.3

Idle →

WaitingStart → Releasing → Returning → Shaking



2

KY-008

8

(
)

8

2

250

3.4

3	Wi-Fi
	5 3
1.2	PWM
	MISSED 30

1-3

4

1 MedicineA x2
 2 MedicineB x1
 3 MedicineC x1
 Next 08:00 2 items

- DISPENSED DISPENSED:

- MISSED:

- SKIPPED:

- FAILED:

- Dry Run: dry run

3.6

Wi-Fi

QR

192.168.4.1

NVS

12

NVS

RTC

PILLBOX_CALIBRATED

PILLBOX_REAL_HARDWARE

IR

4

4.1

tests

25

2569

unittest discover

13

24.3

C++

	2	
	1	
	1	
	1	
LCD	1	
	1	
	2	
	1	
	1	
	2	
	13	

4.2

IR	
	LCD
Wi-Fi	
QR	

1					
2					
3					
4					
5					

$$=$$
$$\times 100$$

$$\div$$

4.3

ESP32	1	168	168
Adabter 5V 5A	1	280	280
DS1307	1	45	45
LCD 16×2	1	73	73
LCD 20×4	1	123	123
	3	91	273
	3	32	96
DRV8833	2	40	80
Laser Head Transmitter Sensor + Laser receiver module	3	50	150
Buzzer	1	7	7
	1	42	42
	1	9	9
Push Button Switch Module 3 3	1	22	22
LED Module 3 traffic light	1	12	12
	1	100	100
	1	398	398
	12	7	84
	1	160	160
google cloud	1	-	
			2,122

(NRE Cost)

$$\begin{array}{rclcl}
 - & 20 & \times 150 & = 3,000 & \\
 - & & 20 & \times 150 & = 3,000 \\
 - & 10 & \times 150 & = 1500 &
 \end{array}$$

$$\mathbf{NRE} = 3,000 + 3,000 + 1,500 = 7,500$$

$$= 2,122 (\quad) + 7,500$$

$$(\mathbf{NRE}) = 9,622$$

(Recurring Cost)

$$\begin{array}{rcl}
 \text{Google Cloud:} & 1 & 3 \\
 & 6 & 30 \\
 70 & / &
 \end{array}$$

5

5.1

ESP32

13

5.2

12

5.3

IR

